

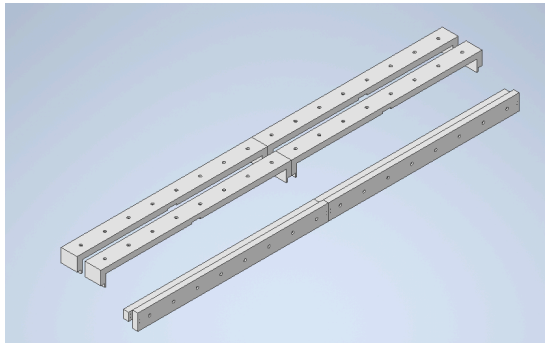
MyDesign Portfolio

Element F: Consideration of Design Viability

Template Instructions This template is designed for technical writing and succinctness. Replace placeholders with precise details from your project, and ensure your responses are backed by credible evidence to fully meet the rubric's criteria.

F1. Design Viability Judgment

- **Design Overview:** Briefly outline your proposed design.



All the spools will be stored on the top and side shelves on vertical and horizontal pegs organized by size, color, or whatever criterion Mr. L wants it to be organized by. The top shelf will be built from plywood and laid across the existing scaffold, which is 144 in x 21 in, so the actual plywood will need to be about an inch smaller on each side. Vertical pegs will have a diameter of 1 inch and will be spaced 9 inches apart, making a grid of 16 x 2 vertical pegs on the shelf, each peg is 16 inches tall. The vertical pegs will serve as long term storage that can store the bulk of the spools. The horizontal pegs will serve as short term storage to allow students to have easy access to the spools to swap out with the existing spools. There will be one horizontal shelf compared to two vertical shelves. The pegs will be 1.5 inches in diameter.

- **Viability Evaluation:** Assess the viability of your design. Provide a realistic judgment backed by evidence.
 - **Evidence:** List evidence supporting viability (e.g., prototype testing, cost analysis, expert opinions).
 - **Justification:** Concisely link each piece of evidence to specific design requirements and practical applications.

In discussion with Mr. L, he told us that he liked our design and saw it as an extremely viable solution to the filament storage problem, specifically surrounding

the design requirements of easy organization, and ability to store 200 spools effectively (see element J). We did a full cost analysis in element D, where we determined that we would need PVC pipe, wood, screws and washers, and 3D printer filament, coming out at a total cost of \$55.44, which is very reasonable for practical application and around the \$50 requested. In element I, we can see that our prototype passed every test, meeting every design requirement, proving itself to be effective and a good design to build fully. In total, the design is very viable, and although individual parts might be able to be tweaked a bit, on the whole it is an effective and realistic solution.

Risk Assessment

- **Identified Risks:** List key risks related to your design (technical, financial, operational).
 - **Mitigation Strategies:** Outline strategies to mitigate these risks with supporting rationale. Reference expert advice or credible data as necessary.
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- The shelf or pegs may not support the weight of the spools, leading to potential collapse.
 - Mitigation: thorough testing of weight-bearing capacity for prototype, calculating the load bearing capacity of the wood.
 - Actual costs may exceed the estimated budget due to unforeseen expenses.
 - Mitigation: fully think through all the steps of building the final product, carefully figure out the price for everything involved, and create an accurate budget.
 - There could be difficulty in installing the shelf and pegs correctly.
 - Mitigation: completely measure the dimensions of the steel shelf beforehand and implement them into the design completely.

Expert Feedback

- **Consultations:** Summarize expert feedback on your design’s viability.

3D printer filament storage

C

Carter Stone <carterstone2005@gmail.com>
to novafablab

Mar 21, 2024, 12:36 PM (13 days ago)

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Dear NOVA Fab Labs,

We are writing this email to ask a couple questions about your 3D printer filament storage. We are students at Marshall high school, and I am working on a design project in my STEM engineering class. Our problem to solve is a 3D printer filament storage system for another STEM lab and teacher in the school. We were curious what your solution to this problem has been at the NOVA Fab Labs, and what methods you have utilized to negate this issue. Thank you so much for your time, and if you have any questions about our design, or qualifiers about the design problem, feel free to reach out.

Sincerely
-Carter Stone



Sewell, Richard A. <rsewell@nvcc.edu>
to ms, novafablab

Mar 26, 2024, 5:30 PM (8 days ago)

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Hey Carter,

Interesting project, and I'd be happy to lend my input.

We currently use a handful of different manufacturers for our spools.

- Most are Stratsys ABS and QSR support spools for our large F-Series printers. These are sealed on shipment, and we make it a point not to unseal them until absolutely necessary – once we have rolls unsealed and not being used in the machines - we keep them in vacuum sealed bags with desiccant packs and humidity meters.
- Our PLA is for our Prusa MK4's, and we are less finicky with this material – partly b/c we go through spools quickly, and also because those machines are very forgiving when it comes to printing slightly wet filament.
- The Markforged machines come with their own dry boxes, so the nylon onyx material stays very dry – in fact one spool has been in there for more than a year and is still good to go – of course these dry boxes contain silica beads as well which we can check for color changes.

Hope that helps – happy to answer additional questions should you need it.

- Richard

Richard Sewell

NOVA SySTEMic
Fabrication Lab Coordinator

- **Feedback Implementation:** Describe adjustments made based on expert feedback, ensuring responses remain technical and focused.

From the expert review we got a good idea of what solution to filament storage they use at NOVA Fab Labs, and we adapted our design accordingly. We kept the actual storage system very close to the 3D printers for the convenience of everyone in the classroom in the design. This decision was based on how they use boxes close to the printers at NOVA Fab Labs, and the convenience that it brings. We switched to dowels/pipes that can be screwed into the shelf using 3D printed connections in the design, also for ease of access which was emphasized in the feedback of the expert reviewer and what they do in their labs. We switched from wooden dowels to cheaper and lighter PVC, which will work better for our design in this context, and which is validated by the use of the cheapest possible materials for storage in the labs.